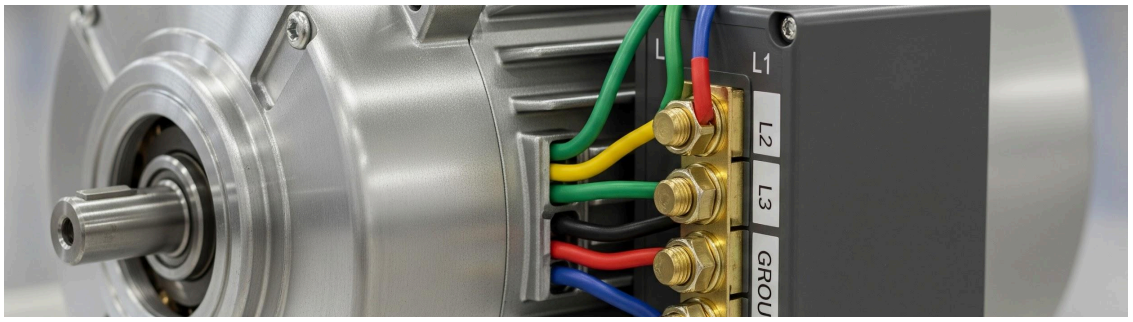

Electrical Machine Data Sheet

Machine Name/Type : Three Phase Squirrel Cage Induction Motor

Name plate Details

The rating section details the machine's operational capacity and design limits.

Parameter	Value	Unit
Power/Output	1	HP
Voltage	415	V
Amps	7.5	A
Frequency	50	Hz
Phase	3	Phase
Speed	1430	RPM

Precautions

I. Ensure the supply is **415 V, 3-phase AC** before starting.

II. Check the correct **phase sequence** to avoid reverse rotation.

III. Do not overload the motor beyond rated capacity.

IV. Ensure proper earthing of the motor body.

V. Use appropriate starter (DOL / Star-Delta) based on motor rating.

VI. Ensure proper cooling and ventilation during operation.

VII. Do not touch rotating parts while the motor is running.

VIII. Monitor current, temperature, and noise continuously.

IX. Avoid frequent starting and stopping.

X. Switch OFF immediately if abnormal vibration, noise, or overheating occurs.

Working Principle of Three Phase Squirrel Cage Induction Motor

When a three-phase AC supply is applied to the stator, it produces a **rotating magnetic field** at synchronous speed.

This rotating field cuts the rotor conductors, inducing current in the squirrel cage rotor according to **Faraday's law of electromagnetic induction**.

The interaction between rotor current and stator magnetic field produces **Lorentz force**, generating torque.

The rotor accelerates and runs at a speed slightly less than synchronous speed, known as **slip**.

Working Characteristics

I. Nearly constant speed under normal load.

II. Good efficiency and power factor.

III. Moderate starting torque.

IV. Simple and rugged construction.

V. Low maintenance due to absence of brushes and slip rings.

VI. High reliability and long service life.

Applications

1. Pumps

2. Fans and blowers

3. Compressors
4. Conveyors
5. Machine tools
6. Elevators
7. Crushers
8. Rolling mills
9. HVAC systems
10. Industrial drives